Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 13: Linearity, Thévenin and Norton Equivalents

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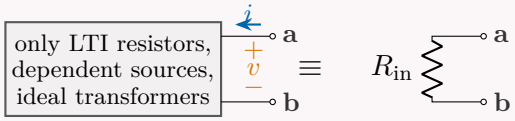
Colour code used in these notes and in the slides			
current i	voltage v, node voltage e	a Δ	a Y

Motivation. Previously, we simplified circuits with equivalent one-ports: series and parallel connections and source transformation. A one-port with dependent sources is not a series–parallel connection of resistors. The first part finds its one equivalent resistor: the **input resistance**. The second part shows that an LTI resistive circuit is **linear** in its independent sources; **superposition** then solves a circuit with several sources one source at a time. The last part adds independent sources inside the one-port: the **Thévenin and Norton** equivalents and the **maximum power transfer** to a load. A take-home part (appendix) gives the **Δ–Y transform** for three resistors that are neither in series nor in parallel.

1. The input resistance of a one-port

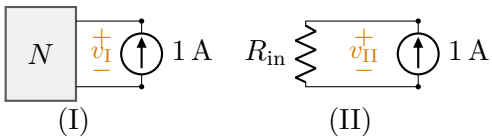
Input resistance

Let a one-port N contain only LTI resistors, LTI dependent sources and ideal transformers, with every control variable inside N. Then N is equivalent to a single LTI resistor R_{in}, the **input resistance** of N: $v = R_{\text{in}}\, i$ at its port.

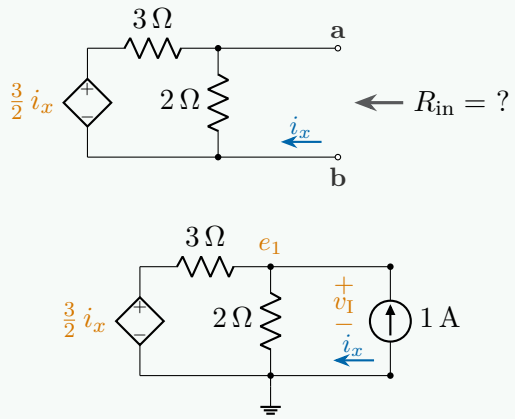


With dependent sources, series and parallel rules are not enough to find R_{in}. A **test source** finds it. Connect a 1 A current source to the port of N (circuit I) and to R_{in} (circuit II). Since $N \equiv R_{\text{in}}$, the port voltages are equal: $v_{\text{I}} = v_{\text{II}} = R_{\text{in}} \cdot 1\,\text{A}$, so

$$R_{\text{in}} = \frac{v_{\text{I}}}{1\,\text{A}}: \quad \text{solve circuit (I).}$$



Example 13.1 — an input resistance



Question. Find R_{in}.
Answer. Connect a 1 A test source and take the bottom node as the ground. KCL at node 1:

$$\frac{e_1 - \frac{3}{2}\,i_x}{3} + \frac{e_1}{2} - 1 = 0.$$

The test current enters at a and leaves at b, so $i_x = -1\,\text{A}$. First plug it in:

$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; \frac{5}{6}\,e_1 = \frac{1}{2} \;\Rightarrow\; e_1 = \frac{3}{5}\,\text{V}.$$

So $v_{\text{I}} = e_1 = \frac{3}{5}\,\text{V}$ and $R_{\text{in}} = \frac{3}{5}\,\Omega$.

With the dependent source replaced by a short circuit, R_{in} would be $3 \parallel 2 = \frac{6}{5}\,\Omega$: the dependent source changes the input resistance.

2. Linearity

In an LTI resistive circuit, the independent sources are the **inputs**. Collect the voltages of the independent voltage sources and the currents of the independent current sources in the **input vector** $\mathbf{u} = [\,u_1 \;\cdots\; u_m\,]^T$. An **output** y is any branch voltage or current.

Linearity

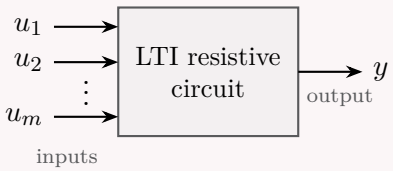
Input u_j alone (the other inputs zero): the output is

$$y_j = k_j\,u_j,$$

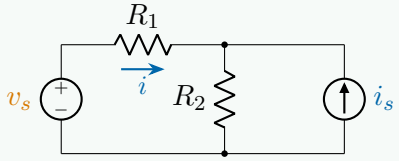
with k_j a constant set by the resistors and the dependent-source gains.
All inputs: the output is the sum of these parts,

$$y = y_1 + y_2 + \cdots + y_m = k_1 u_1 + \cdots + k_m u_m = \mathbf{k}\,\mathbf{u},$$

with the row vector $\mathbf{k} = [\,k_1 \;\cdots\; k_m\,]$.



Example 13.2 — the row vector k



Question. The inputs are $\mathbf{u} = [\,v_s \;\; i_s\,]^T$. For the output $y = i$, find k_1, k_2 with $i = i_1 + i_2$, $i_1 = k_1 v_s$, $i_2 = k_2 i_s$.
Answer. KVL around the left mesh, with $i + i_s$ down through R_2:

$$-v_s + R_1 i + R_2(i + i_s) = 0,$$

$$i = \underbrace{\frac{1}{R_1 + R_2}\,v_s}_{i_1} \underbrace{- \frac{R_2}{R_1 + R_2}\,i_s}_{i_2},$$

so $k_1 = 1/(R_1 + R_2)$ and $k_2 = -R_2/(R_1 + R_2)$.

The implication of linearity

For input vectors $\mathbf{u}$, $\mathbf{w}$ and numbers α, β:

$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w}).$$

3. Superposition

Split the input vector into one source at a time. With two sources,

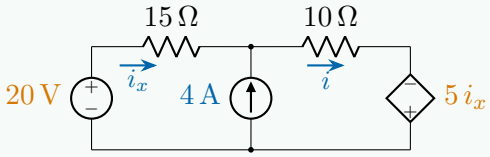
$$y\left(\begin{bmatrix} v_s \\ i_s \end{bmatrix}\right) = y\left(\begin{bmatrix} v_s \\ 0 \end{bmatrix}\right) + y\left(\begin{bmatrix} 0 \\ i_s \end{bmatrix}\right):$$

the output is the sum of the outputs with one independent source at a time. The other sources are **killed**.

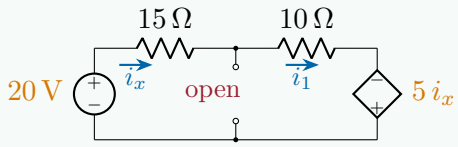
Killing a source

Killing an independent voltage source ($v_s = 0$) replaces it with a **short circuit**. Killing an independent current source ($i_s = 0$) replaces it with an **open circuit**. Do **not** kill dependent sources: they are not inputs.

Example 13.3 — superposition



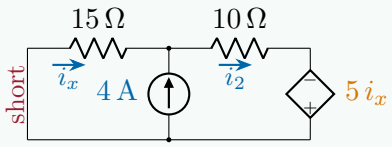
Question. Find i by superposition.
Answer. $i = i_1 + i_2$: i_1 with the 4 A source killed, i_2 with the 20 V source killed.



Kill the current source (open). One loop, $i_x = i_1$:

$$-20 + 15\,i_x + 10\,i_1 - 5\,i_x = 0$$

$$\Rightarrow\ 20\,i_1 = 20\ \Rightarrow\ i_1 = 1\,\text{A}.$$



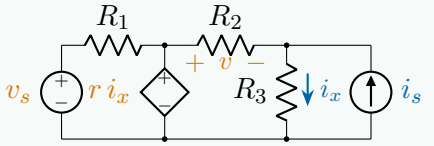
Kill the voltage source (short). KCL: $i_x = i_2 - 4$:

$$15\,i_x + 10\,i_2 - 5\,i_x = 0$$

$$\Rightarrow\ 20\,i_2 = 40\ \Rightarrow\ i_2 = 2\,\text{A}.$$

So $i = i_1 + i_2 = 3\,\text{A}$. The dependent source $5\,i_x$ stays in both circuits.

Example 13.4 — three experiments



	v_s	i_s	v
Exp. 1	12 V	0	v_1
Exp. 2	−6 V	6 A	v_2
Exp. 3	4 V	4 A	v_3

Question. Three experiments are made on the circuit. Express v_3 in terms of v_1 and v_2.
Answer. By linearity, $v = \mathbf{k}\,[\,v_s \;\; i_s\,]^T$ for some $\mathbf{k}$. Write the third input with the first two:

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \ \Rightarrow\ 6\beta = 4,\quad 12\alpha - 6\beta = 4\ \Rightarrow\ \alpha = \beta = \tfrac{2}{3}.$$

Then $v_3 = \alpha\,v_1 + \beta\,v_2 = \frac{2}{3}\,v_1 + \frac{2}{3}\,v_2$. Equivalently, $\mathbf{k} = [\,v_1 \;\; v_2\,]\,\left[\begin{smallmatrix} 12 & -6 \\ 0 & 6 \end{smallmatrix}\right]^{-1}$. No element value is needed.

4. Thévenin and Norton equivalent circuits

With independent sources inside, a one-port is no longer one resistor: it needs one more element, a source. Whatever is connected to the port, a load (a lamp, a motor) or another circuit, then sees only a source and a resistor, found once.

Thévenin's theorem

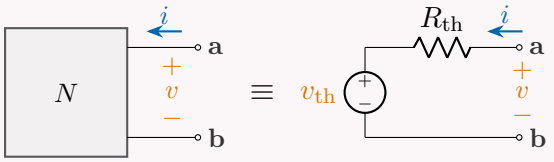
Let N be an LTI resistive one-port, possibly with independent sources, such that

(A1) every dependent source in N has its control variable in N;

(A2) the port is both voltage- and current-controlled.

Then there are v_{th} and R_{th} with N equivalent to the circuit on the right:

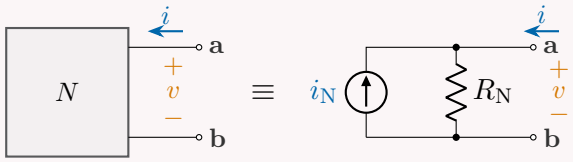
$$v = R_{\text{th}}\, i + v_{\text{th}}.$$



Norton's theorem

Under the same conditions (A1), (A2), there are i_{N} and R_{N} with N equivalent to the circuit on the right:

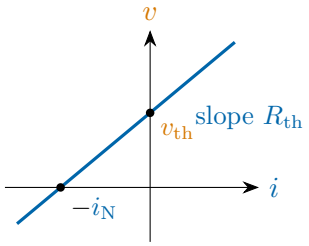
$$v = R_{\text{N}}\, i + R_{\text{N}}\, i_{\text{N}}.$$



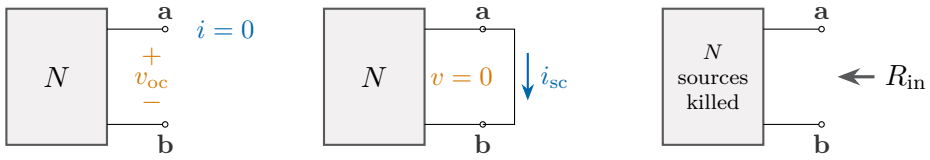
Thévenin and Norton describe the same line. Both characteristics are the same straight line in the i–v plane, so

$$R_{\text{th}} = R_{\text{N}} \qquad \text{and} \qquad v_{\text{th}} = R_{\text{th}}\, i_{\text{N}}.$$

Moving between the two is the source transformation (previously). The line meets the v axis at v_{th} and the i axis at $-i_{\text{N}}$; its slope is R_{th}.



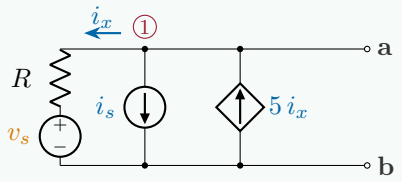
5. Computing v_{th}, i_{N} and R_{th}



1. **Open circuit.** With the port open, $i = 0$ and the Thévenin characteristic gives $v = v_{\text{th}}$: $v_{\text{th}} = v_{\text{oc}}$.
2. **Short circuit.** With the port shorted, $v = 0$, so $i = -i_{\text{N}}$. The current through the short, from a to b, is $i_{\text{sc}} = -i$: $i_{\text{N}} = i_{\text{sc}}$.
3. $\boldsymbol{R_{\text{th}}}$**.** Once v_{th} and i_{N} are known, $R_{\text{th}} = v_{\text{th}}/i_{\text{N}} = v_{\text{oc}}/i_{\text{sc}}$. Another way: kill all independent sources in N (voltage source: short, current source: open) and compute the input resistance, $R_{\text{th}} = R_{\text{in}}$. Killing the independent sources sets $v_{\text{th}} = 0$ and $i_{\text{N}} = 0$; only R_{th} is left. The second way also works when $v_{\text{oc}} = i_{\text{sc}} = 0$.

6. Examples

Example 13.5 — Thévenin and Norton equivalents



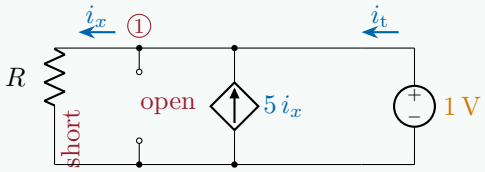
Find the Thévenin and Norton equivalents of the one-port a–b.

Step 1: open circuit. KCL at node 1 with $i = 0$:

$$i_x + i_s - 5\,i_x = 0 \;\Rightarrow\; i_x = \tfrac{1}{4}\,i_s,$$
$$v_{\text{th}} = v_{\text{oc}} = R\,i_x + v_s = \tfrac{R}{4}\,i_s + v_s.$$

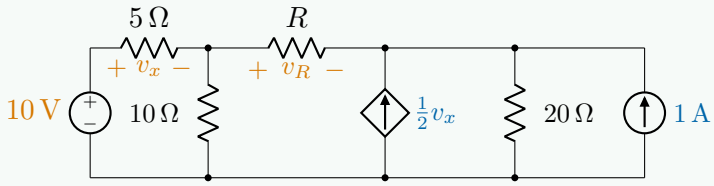
Step 2: short circuit. Node 1 is at 0 V, so $i_x = -v_s/R$. KCL at node 1: $i_x + i_s - 5\,i_x + i_{\text{sc}} = 0$, so $i_{\text{N}} = i_{\text{sc}} = -i_s + 4\,i_x = -i_s - \frac{4}{R}\,v_s$.

Step 3: R_{th}. $R_{\text{th}} = v_{\text{th}}/i_{\text{N}} = -R/4$. It is negative: the dependent source supplies power.

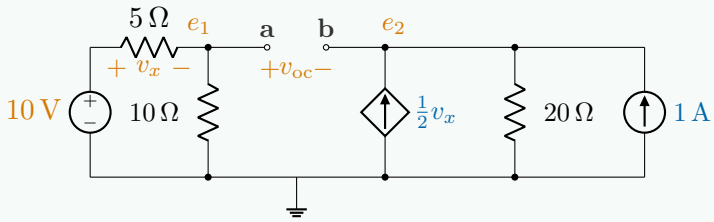


Another way. Kill v_s (short) and i_s (open) and connect a 1 V test source. Then $i_x = 1/R$, and KCL at node 1 gives $i_x - 5\,i_x - i_t = 0$, so $i_t = -4/R$ and $R_{\text{th}} = R_{\text{in}} = 1\,\text{V}/i_t = -R/4$, as expected.

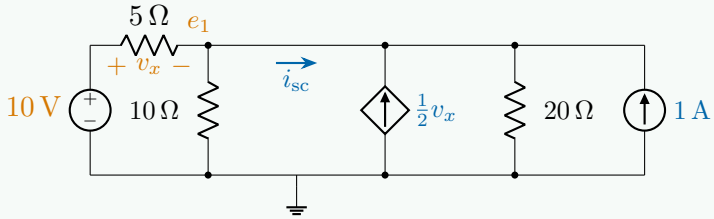
Example 13.6 — a load to design



The load is R. Find the Thévenin equivalent seen by R. Then find R such that $v_R = -7\,\text{V}$.



Open circuit (remove R). Node 1: $\frac{e_1-10}{5} + \frac{e_1}{10} = 0$, so $e_1 = \frac{20}{3}\,\text{V}$ and $v_x = 10 - e_1 = \frac{10}{3}\,\text{V}$. Node 2: $-\frac{1}{2}v_x + \frac{e_2}{20} - 1 = 0$, so $e_2 = \frac{160}{3}\,\text{V}$ and $v_{\text{th}} = e_1 - e_2 = -\frac{140}{3}\,\text{V}$.



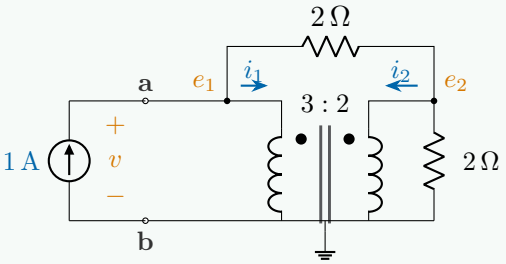
Short circuit. One node e_1: $\frac{e_1-10}{5} + \frac{e_1}{10} - \frac{1}{2}v_x + \frac{e_1}{20} - 1 = 0$. Plug $v_x = 10 - e_1$ first: $\frac{17}{20}\,e_1 = 8$, $e_1 = \frac{160}{17}\,\text{V}$. Left of the short, $\frac{e_1-10}{5} + \frac{e_1}{10} + i_{\text{sc}} = 0$, so $i_{\text{N}} = i_{\text{sc}} = -\frac{14}{17}\,\text{A}$.

So $R_{\text{th}} = v_{\text{th}}/i_{\text{N}} = \frac{170}{3}\,\Omega$. With R connected to this Thévenin equivalent, a voltage divider gives

$$v_R = \frac{R}{R + \frac{170}{3}}\left(-\frac{140}{3}\right) = -7 \;\Rightarrow\; 140\,R = 21\,R + 1190 \;\Rightarrow\; R = 10\,\Omega.$$

Check. Killing the sources (the dependent one stays) also gives $R_{\text{th}} = \frac{10}{3} + \frac{160}{3} = \frac{170}{3}\,\Omega$.

Example 13.7 — an input resistance with an ideal transformer

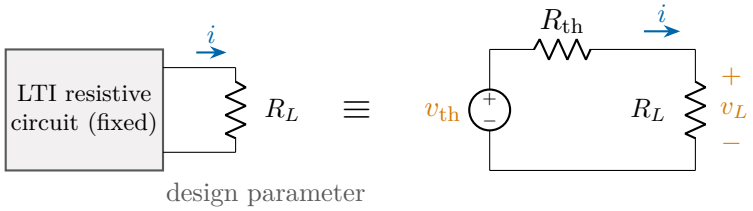


Find R_{in} at the port a–b. There is no independent source inside, so the Thévenin equivalent is R_{in} alone; connect a 1 A test source. The ideal transformer $3:2$ (previously): $e_1/3 = e_2/2$ and $3\,i_1 + 2\,i_2 = 0$.

Answer. Node 1: $-1 + i_1 + \frac{1}{2}(e_1 - e_2) = 0$; node 2: $i_2 + \frac{1}{2}e_2 + \frac{1}{2}(e_2 - e_1) = 0$. Plug $e_2 = \frac{2}{3}\,e_1$ and $i_2 = -\frac{3}{2}\,i_1$: node 2 gives $i_1 = e_1/9$, and node 1 gives $\frac{5}{18}\,e_1 = 1$, so $e_1 = \frac{18}{5}$ V. Then $v = e_1$ and $R_{\text{in}} = \frac{18}{5}\,\Omega$.

7. Maximum power transfer

In many applications a fixed circuit should deliver as much power as possible to a load: an amplifier drives a speaker, an antenna feeds a receiver. The circuit is fixed; the load resistance R_L is the design parameter.



design parameter

7.1. The power delivered to the load

Replace the fixed circuit by its Thévenin equivalent; v_{th} and R_{th} do not depend on R_L. Then

$$i = \frac{v_{\text{th}}}{R_{\text{th}} + R_L}, \qquad P_L = R_L\,i^2 = \frac{R_L\,{v_{\text{th}}}^2}{(R_{\text{th}} + R_L)^2},$$

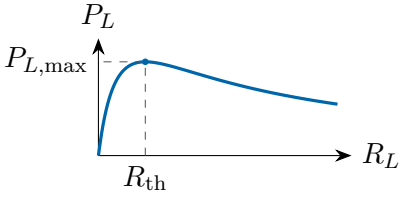
a function of R_L only. Two extremes give no power: $R_L = 0$ is a short circuit, so $v_L = 0$; $R_L \to \infty$ is an open circuit, so $i = 0$. The maximum is in between.

7.2. The maximum

Differentiate with respect to R_L:

$$\frac{dP_L}{dR_L} = {v_{\text{th}}}^2\,\frac{(R_{\text{th}} + R_L) - 2R_L}{(R_{\text{th}} + R_L)^3} = {v_{\text{th}}}^2\,\frac{R_{\text{th}} - R_L}{(R_{\text{th}} + R_L)^3}.$$

The derivative is positive for $R_L < R_{\text{th}}$ and negative for $R_L > R_{\text{th}}$, so P_L has its maximum at $R_L = R_{\text{th}}$.



Maximum power transfer

A load R_L receives the maximum power from a circuit when $R_L = R_{\text{th}}$:

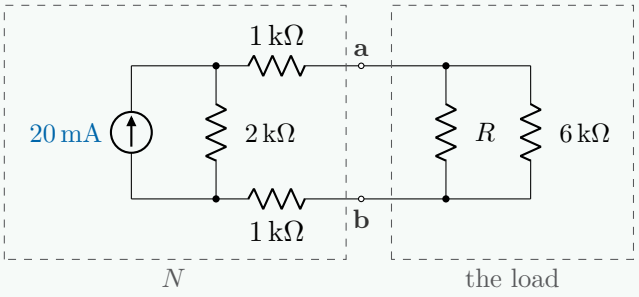
$$P_{L,\max} = \frac{{v_{\text{th}}}^2}{4R_{\text{th}}} = \frac{1}{4}\,R_{\text{N}}\,{i_{\text{N}}}^2.$$

7.3. At maximum power

With $R_L = R_{\text{th}}$, the load voltage is $v_L = \frac{1}{2}\,v_{\text{th}}$ and the current is $i = v_{\text{th}}/(2R_{\text{th}})$. In the Thévenin equivalent the source supplies $v_{\text{th}}\,i = {v_{\text{th}}}^2/(2R_{\text{th}})$: half of it reaches the load and half is dissipated in R_{th}. This statement is about the equivalent; the power dissipated inside the original circuit can be different.

7.4. An example

Example 13.8 — maximum power



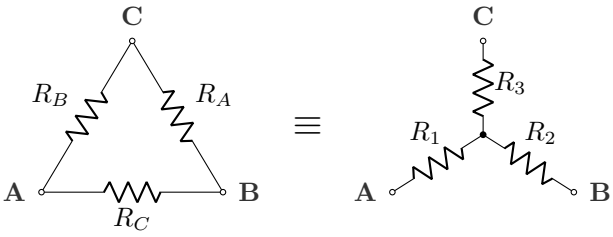
Find R so that the maximum power is delivered to the load, and find this power.
Thévenin equivalent of N. Open circuit: $v_{\text{th}} = 20\,\text{mA} \times 2\,\text{k}\Omega = 40\,\text{V}$. Kill the source (open): $R_{\text{th}} = 1 + 2 + 1 = 4\,\text{k}\Omega$.
The load. $R \parallel 6\,\text{k}\Omega = R_{\text{th}} = 4\,\text{k}\Omega$ gives $R = 12\,\text{k}\Omega$, and $P_{L,\max} = {v_{\text{th}}}^2/(4R_{\text{th}}) = 0.1\,\text{W}$.

Summary

In the circuit	What to do
resistors, dependent sources	a 1 A test source: $R_{\text{in}} = v/1\,\text{A}$
a linear circuit	$y = \mathbf{k}\,\mathbf{u}$; $y(\alpha\mathbf{u} + \beta\mathbf{w}) = \alpha\, y(\mathbf{u}) + \beta\, y(\mathbf{w})$
several independent sources	superposition; killed v_s: short, killed i_s: open; dependent sources stay
independent sources inside a load R_L to choose	$v_{\text{th}} = v_{\text{oc}}$, $i_{\text{N}} = i_{\text{sc}}$, $R_{\text{th}} = R_{\text{N}} = v_{\text{oc}}/i_{\text{sc}}$ or R_{in} with the sources killed maximum power for $R_L = R_{\text{th}}$: $P_{L,\max} = {v_{\text{th}}}^2/(4R_{\text{th}})$

A. Take-home: the Δ–Y transform

Three resistors may form a Δ or a Y between three terminals A, B, C. In the Δ, R_A is opposite A (between B and C), R_B is opposite B and R_C is opposite C. In the Y, R_1, R_2, R_3 are at A, B, C.



Δ to Y. Given R_A, R_B, R_C, find R_1, R_2, R_3. If the two are equivalent, the resistance between any two terminals, with the third one open, is the same. Between A and B, the Δ is R_C in parallel with $R_A + R_B$, and the Y is $R_1 + R_2$:

$$\frac{R_C(R_A+R_B)}{R_A+R_B+R_C} = R_1+R_2 \quad (1), \qquad \frac{R_A(R_B+R_C)}{R_A+R_B+R_C} = R_2+R_3 \quad (2),$$

$$\frac{R_B(R_C+R_A)}{R_A+R_B+R_C} = R_3+R_1 \quad (3).$$

Half of the sum of (1), (2) and (3) gives

$$R_1+R_2+R_3 = \frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \qquad (4),$$

and (4)−(2), (4)−(3), (4)−(1) give the result.

Δ to Y

$$R_1 = \frac{R_BR_C}{R_A+R_B+R_C}, \qquad R_2 = \frac{R_CR_A}{R_A+R_B+R_C}, \qquad R_3 = \frac{R_AR_B}{R_A+R_B+R_C}$$

The Y resistor at a terminal is the product of the two Δ resistors at that terminal, divided by the sum of the Δ resistors.

Y to Δ. Given R_1, R_2, R_3, find R_A, R_B, R_C. From the result above,

$$S = R_1R_2+R_2R_3+R_3R_1 = \frac{R_AR_BR_C\,(R_A+R_B+R_C)}{(R_A+R_B+R_C)^2} = \frac{R_AR_BR_C}{R_A+R_B+R_C},$$

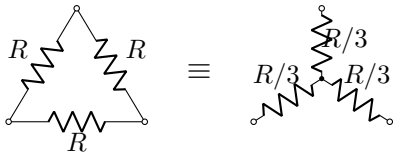
$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A+R_B+R_C} = S.$$

Y to Δ

$$R_A = \frac{S}{R_1}, \qquad R_B = \frac{S}{R_2}, \qquad R_C = \frac{S}{R_3}, \qquad S = R_1R_2+R_2R_3+R_3R_1$$

A Δ resistor is the sum of the products of the Y resistors, divided by the Y resistor opposite to it.

Equal resistors. A Δ of three equal resistors R is equivalent to a Y of three resistors $R \cdot R/(3R) = R/3$. Conversely, a Y of three resistors R is equivalent to a Δ of three resistors $3R$.

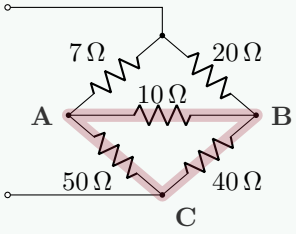


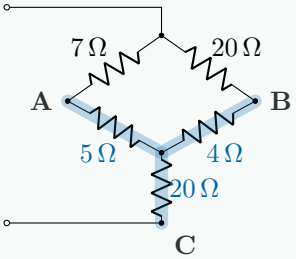
Exercise. Write both transforms with conductances, $G_\square = 1/R_\square$. *Answer:* Y to Δ becomes $G_A = G_2G_3/(G_1+G_2+G_3)$, and so on; it has the form of Δ to Y with resistances.

An example: a bridge

Example 13.9 — a bridge

Question. Find the resistance R between the two terminals. No two resistors are in series or in parallel.





Answer. Transform the Δ A–B–C into a Y: $R_C = 10\,\Omega$, $R_B = 50\,\Omega$, $R_A = 40\,\Omega$, and $R_A + R_B + R_C = 100\,\Omega$. So

$$R_1 = \frac{50 \cdot 10}{100} = 5\,\Omega, \qquad R_2 = \frac{10 \cdot 40}{100} = 4\,\Omega, \qquad R_3 = \frac{40 \cdot 50}{100} = 20\,\Omega.$$

In the new circuit, $7 + 5 = 12\,\Omega$ and $20 + 4 = 24\,\Omega$ are in parallel, $12 \parallel 24 = 8\,\Omega$, and this is in series with $20\,\Omega$:

$$R = 8 + 20 = 28\,\Omega.$$

Check. With node voltages, a 1 A test source between the terminals gives a terminal voltage of 28 V. ✓